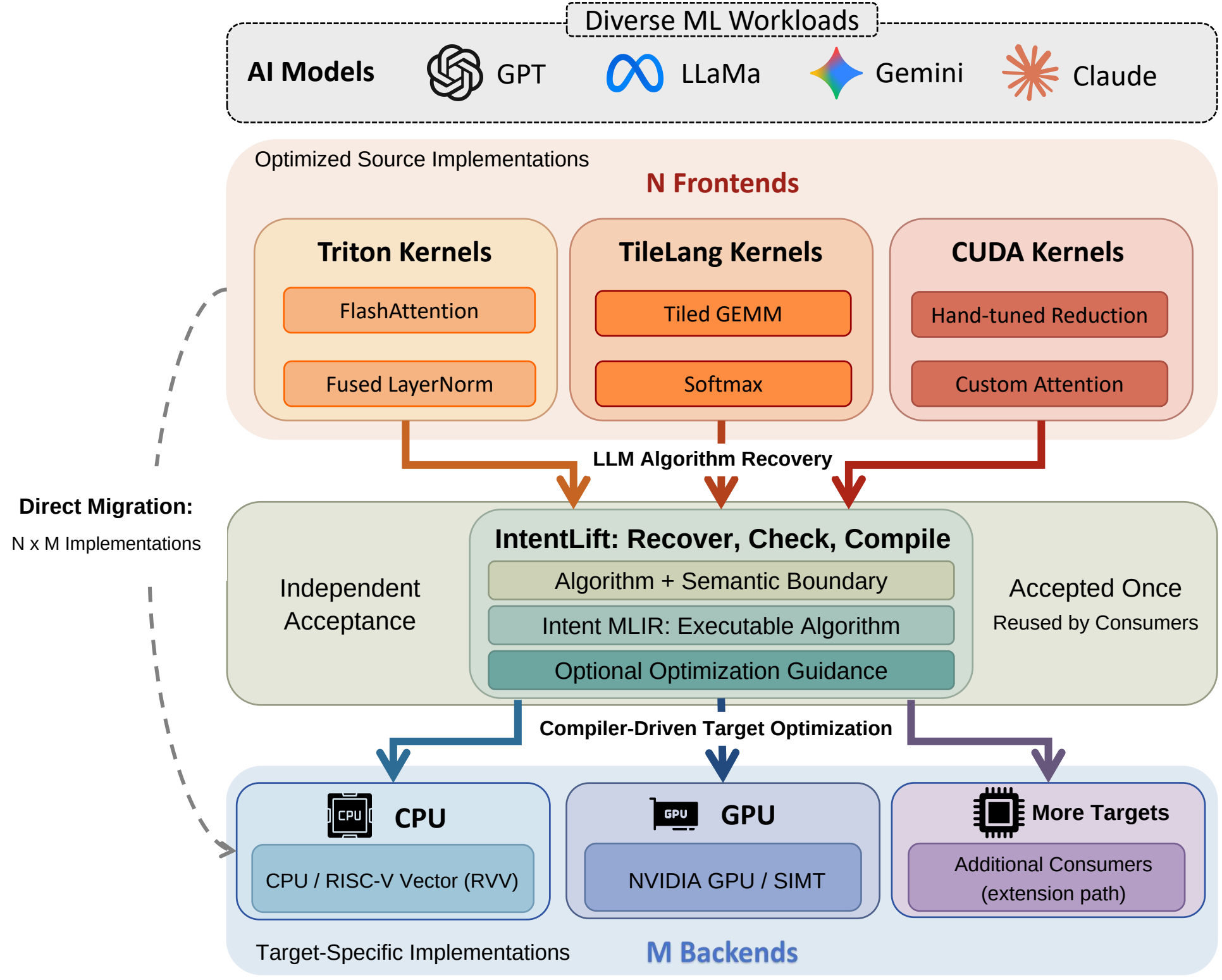

Diverse ML Workloads
AI Models
GPT
LLaMa
Gemini
Claude
Optimized Source Implementations
N Frontends
Triton Kernels
FlashAttention
Fused LayerNorm
TileLang Kernels
Tiled GEMM
Softmax
CUDA Kernels
Hand-tuned Reduction
Custom Attention
LLM Algorithm Recovery
Direct Migration:
N x M Implementations
Independent Acceptance
IntentLift: Recover, Check, Compile
Algorithm + Semantic Boundary
Intent MLIR: Executable Algorithm
Optional Optimization Guidance
Accepted Once
Reused by Consumers
Compiler-Driven Target Optimization
CPU
CPU / RISC-V Vector (RVV)
GPU
NVIDIA GPU / SIMT
More Targets
Additional Consumers (extension path)
Target-Specific Implementations
M Backends